

sahil sharma

AI and ML Engineer

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AI and Machine Learning Engineer with expertise in developing innovative algorithms and data-driven solutions. Lead Generative AI Engineer with 6 years experience in python, GenAI, LLM, MLOPS, DevOps, machine learning, data science, predictive modeling, and statistical analysis to drive actionable insights. Proficient in Python, TensorFlow, and data visualization tools, with experience in designing and deploying scalable AI models. Adept at leveraging big data technologies and applying advanced analytics to solve complex business challenges. Strong background in data preprocessing, feature engineering, and model optimization to enhance performance. Committed to continuous learning and staying current with emerging AI trends and technologies to deliver impactful results in diverse industries.

SKILLS

- Deep Learning
- Artificial Intelligence
- Data Extraction
- Java
- Data Engineering
- Matplotlib
- Numpy
- Cloud Computing
- Tensorflow
- Huggingface
- Apache spark
- Tfs / Github
- Azure Data Factory
- Databricks Engineer
- Celery
- Statistics
- Tableau
- Data Visualization
- Python
- Data Science Consultant
- Seaborn
- Pandas
- MySQL
- Keras
- Scrapy
- Docker
- Agentic Ai
- Retrieval Augmented Generation
- Snowflake DB
- Redis
- Data Mining
- Microsoft Power Bi
- Power Bi
- Data Analytics
- Data Science Business Analyst
- Sckit learn
- Azure DevOps
- Large Language Model
- Pytorch
- Apache Tomcat
- Kubernetes
- AWS
- MongoDB
- Rabbitmq

CERTIFICATION

- Sanyss technologies

WORK EXPERIENCE

AI and ML Engineer - PhonePe

Jul 2020 - Present

Role summary

Leading AI quality standards with LLM , NLP pipeline and generative AI workflow from design to production end. Utilized advanced AI and machine learning techniques to develop and optimize predictive models.

Azure cloud deployment and train Model on different datasets.

CORE SKILLS

1. Software Testing Skills

Manual testing (functional, regression, sanity, smoke, UAT, blackbox, retesting, cross browser, Compatibility, accessibility)

API Testing (Postman, Rest assured, SoapUI, Swagger)

Automation testing (selenium, webDriver, python, java, TestNG, Maven, Cucumber, Appium)

Performance Testing (JMeter)

Unix commands, Troubleshooting, Defect Tracking

AI and Generative AI Skills

Generative AI Testing, Agentic AI Testing, Prompt Engineering, AI Prompt Testing, Rag Pipeline Testing.

AI / ML Developer — Core Skills

- ◆ Programming & Tools

Python (Primary)

NumPy, Pandas

Scikit-learn

PyTorch, TensorFlow

SQL / MySQL

Git, GitHub

- ◆ Machine Learning

Supervised & Unsupervised Learning

Regression, Classification, Clustering

Feature Engineering & Feature Selection

Model Training, Tuning & Optimization

Cross-validation & Hyperparameter tuning

Model Evaluation (Accuracy, Precision, Recall, F1, ROC-AUC)

- ◆ Deep Learning

Neural Networks

CNNs, RNNs, LSTMs

Transfer Learning

Model fine-tuning

GPU-based training

- ◆ Natural Language Processing (NLP)

Text preprocessing & tokenization

Word embeddings

Text classification & sentiment analysis

Named Entity Recognition (NER)

Question Answering systems

- ◆ Generative AI & LLMs

Large Language Models (LLMs)

Prompt Engineering & Prompt Optimization

RAG (Retrieval Augmented Generation) pipelines

Vector databases & embeddings

Hallucination detection & guardrails

Agentic AI workflows

- ◆ Model Deployment & APIs

FastAPI, Flask

REST API development for ML models

Model inference pipelines

Batch & real-time prediction systems

- ◆ MLOps & Engineering

- End-to-end ML pipelines
- Model versioning
- Model monitoring & drift detection
- CI/CD for ML workflows
- Docker & containerization
- Experiment tracking
 - ◆ Cloud Platforms
- AWS (SageMaker, EC2, S3, Lambda)
- Azure (AI & DevOps exposure)
- Cloud-based training & deployment
 - ◆ Data & Visualization
- Data cleaning & preprocessing
- SQL-based data analysis
- Power BI / dashboards (basic to intermediate)
- Model performance reporting

✓ AI / ML Developer — Roles & Responsibilities

- ◆ Data & Model Development
- Analyze large datasets to extract insights and patterns
- Design, train, and optimize machine learning models
- Perform feature engineering and data preprocessing
- Select appropriate algorithms based on business use cases
 - ◆ Model Evaluation & Optimization
- Evaluate models using appropriate metrics
- Perform hyperparameter tuning and model optimization
- Address overfitting, underfitting, and bias issues
- Improve model accuracy and robustness
 - ◆ AI System Design
- Design end-to-end AI/ML pipelines
- Build NLP and Generative AI solutions
- Implement RAG-based and agentic AI systems
- Integrate models into production systems
 - ◆ Deployment & Production
- Develop APIs for model serving
- Deploy models to cloud environments
- Monitor model performance in production
- Handle model retraining and updates
 - ◆ MLOps & Automation
- Implement CI/CD pipelines for ML models
- Manage model versioning and experiment tracking
- Automate training, testing, and deployment workflows
- Ensure scalability and reliability of AI systems
 - ◆ Collaboration & Communication
- Work closely with product, engineering, and business teams
- Translate business requirements into ML solutions
- Participate in design reviews and technical discussions
- Document models, pipelines, and experiments
 - ◆ Responsible AI & Quality
- Implement ethical AI practices
- Handle data privacy and security
- Detect and mitigate model hallucinations
- Apply guardrails and validation checks

AI/ML Developer with hands-on experience in building, deploying, and monitoring machine learning and generative AI solutions, specializing in NLP, LLMs, and scalable MLOps pipelines.

Data Analyst - Walmart Global Tech

Jul 2015 - May 2016

- Developed a Retrieval-Augmented Generation (RAG) model to address complex customer problems effectively.
- Leveraged advanced data science techniques to enhance model accuracy and customer satisfaction.
- Collaborated with cross-functional teams to integrate the model into customer support workflows, improving response times by approximately 20%.

PROJECTS

Ai chatbot for customer support

6 Months

DOMAIN = AI Chatbot
Client = phonepe india private limited
Role = GenAI engineer

- Developed an AI-powered chatbot to enhance customer support by efficiently handling diverse query types
- Integrated natural language processing (NLP) to improve response accuracy and customer satisfaction
 - Reduced average query resolution time by approximately 30%, improving overall support efficiency

Skills
Python , GenAI, ML, NLP, LLM evaluation, AI risk scoring, AWS, Azure, CI/CD , LLM Prompt validation, API based AI services, SQL, DOCKER, GIT, GITHUB, Automation frameworks, APIs, Agile, Mobile and web testing, Data validation, power Bi, cloud computing. Agentic AI Model development
Azure cloud

Roles and responsibilities

- 1 Validated data driven and AI assistant decision workflows in chat system.
- 2 Ensure accuracy and trustworthy of analytics and recommendations.
- 3 Preformed backed data validation and API automation for financial workflow.
- 5 Collaborated with cross functional teams
- 6 Azure cloud deployment

ETL pipeline project

6 Months

📌 Project Title

Customer Analytics ETL Pipeline for E-commerce Platform

📌 Business Problem

Data comes from multiple sources
Analysts need clean.

📌 Data Sources

Source
Type
Orders API
REST API (JSON)
Customer DB

MySQL
Clickstream
Kafka
S3

ETL Architecture (High Level)

Architecture Flow
Data Ingestion
Batch + Streaming
Raw Storage
Data Lake (Bronze)
Transformation
Spark / SQL
Curated Storage
Data Warehouse (Gold)
Consumption
BI, ML, APIs

Tools & Technologies (Industry-Standard Stack)

◆ Data Ingestion

Kafka
Real-time streaming
Airflow
Batch orchestration
AWS Glue
Serverless ETL
REST APIs
External data

◆ Data Processing

Tool
Use Case
Apache Spark
Large-scale transformation
PySpark
Distributed Python ETL
SQL
Aggregations & joins
Pandas
Small batch jobs

◆ Storage Layer

Layer
Tool
Raw (Bronze)
S3 / ADLS
Cleaned (Silver)
Parquet
Aggregated (Gold)

Redshift / BigQuery / Snowflake

◆ Orchestration & Monitoring

Tool

Role

Airflow

DAG scheduling

◆ DevOps & CI/CD

Git

Version control

Docker

Containerization

Jenkins / GitHub Actions

CI/CD

Customer churn prediction

4 Months

Project Title

Customer Churn Prediction System

📌 Business Problem

Companies lose revenue when customers stop using their service (churn).

The goal is to predict which customers are likely to churn so the business can take preventive action (offers, retention calls, discounts).

📌 Business Impact

Reduce customer loss

Increase customer lifetime value (CLV)

Optimize marketing spend

📌 Use Case Examples

Telecom (postpaid users)

Banking (credit card churn)

SaaS subscriptions

OTT platforms

📌 High-Level Architecture

Flow:

Data Ingestion

Data Cleaning & Feature Engineering

Model Training

Model Evaluation

Deployment

Monitoring

📌 Dataset Description

◆ Typical Features

Category
Examples
Customer Info
Age, Gender, Location
Account Info
Tenure, Contract Type
Usage
Monthly charges, Data usage
Support
Complaints, Call count
Target
Churn (Yes/No)

Data Pipeline (ETL Perspective)

◆ Extract

CSV files

Database (MySQL / Redshift)

Data Lake (S3)

Exploratory Data Analysis (EDA)

🔍 Key Insights

Customers with month-to-month basis.

Credit card fraud detection

4 Months

Real-Time Credit Card Fraud Detection System

🔗 Business Problem

Credit card fraud causes huge financial losses and customer trust issues.

The goal is to detect fraudulent transactions in real time with high recall while minimizing false positives.

🔗 Business Impact

Prevent financial loss

Reduce chargebacks

Improve customer trust

Meet regulatory requirements

Fraud Detection Use Cases

Card-not-present transactions

Unusual spending behavior

Geo-location mismatch

High-velocity transactions

System Architecture (High Level)

Flow:

Transaction ingestion (streaming)

Feature engineering
Fraud prediction
Decision engine
Alerts & actions
Monitoring & retraining

Data Sources

Source
Type
Transaction logs
Kafka streams
User profile
RDBMS
Merchant data
APIs
Historical fraud labels
Data warehouse

Data Pipeline (Streaming + Batch)

◆ Streaming Pipeline (Real-Time)
Kafka → Spark Structured Streaming
Feature generation in milliseconds
Online inference
◆ Batch Pipeline (Offline)
Historical data analysis

Multiple model rag chatting with pdfs containg images and table

6 Months

Project Description:

Retail optimization platform focused on customer experience, analytics, and personalized engagement.

Skills :

Python, Generative AI, ML,NLP, LLM evaluation, AI risk scoring, AWS, Azure, CI/CD, LLM prompt validation, API-based AI services, SQL, AWS, CI/CD, Docker, Git,

Automation Frameworks,

APIs, Agile, Automation, Mobile & Web Testing, APIs, Agile, Automation, Mobile & Web Testing, LLM prompt validation, API-based AI services, Agile, APIs, Data Validation, Power BI, Cloud Computing.

Roles & Responsibilities:

1. Validated recommendation and personalization logic.
2. Performed API and backend data validations.
3. Supported AI-driven analytics validation.
4. Automated regression and integration testing.
5. Collaborated with data engineers and business teams.
6. Supported cloud-based deployments.
7. Ensured performance and scalability validation.

- 8. Prepared dashboards and reports.
- 9. Conducted defect analysis and RCA.

Build multi model using RAG with langchain

12 Months

Environment Setup and Data Extraction

Install necessary libraries: I will need langchain, openai, chromadb (or another vector store), and unstructured for parsing complex documents like PDFs.

Load Documents: Use LangChain's document loaders and the unstructured library to parse documents (e.g., PDFs) into distinct elements like text, tables, and images.

Data Processing and Summarization

Categorize Elements: Separate the extracted data into their respective types: raw text chunks, tables, and images.

Summarize Content: Generate concise text summaries for each element type using an LLM (like GPT-4o mini or GPT-4 Vision).

For images, the LLM analyzes the visual data and provides a descriptive text summary.

For tables, the LLM summarizes the key information and structure.

Indexing and Storage

Create Embeddings: Generate vector embeddings for all the summaries (text, table, and image summaries) using an embedding model (e.g., OpenAI embeddings or a HuggingFace model)

EDUCATION

2027 M.Tech | Computers
Manipal Academy of Higher Education (MAHE)
Grade - 9/10

2020 B.Tech/B.E. | Computer Science
Punjab Technical University (PTU)
Grade - 8/10

2011 12th
Punjab, English
Marks - 70-74.9%

2009 10th
Punjab, English
Marks - 80-84.9%

ADDITIONAL INFORMATION

Languages: English, Punjabi, Hindi